Zomato Restaurant Data Analysis

Week 1 Assignment - Exploratory Data Analysis of the Zomato restaurant dataset using Python (Pandas, NumPy, Matplotlib, Seaborn).

UST Assignment 1 - Documentation Report

---

1. Dataset Overview

The dataset has information about restaurants listed on Zomato, like where they are located, what cuisines they serve, how much it costs, their ratings and how many people voted. It is read from the file **test_zomato.csv**.

The dataset has **9552 rows** and **22 columns**. The main columns I worked with are:

Key columns in the dataset

Column	Description
Restaurant Name	Name of the restaurant
City	City where the restaurant is located
Cuisines	Type of cuisines served
Average Cost for two	Estimated cost for two people
Has Online delivery	Whether online delivery is available
Has Table booking	Whether table booking is available
Price range	Restaurant price category (1 to 4)
Aggregate rating	Overall customer rating (0 to 5)
Rating text	Rating category like Excellent or Average
Votes	Number of customer votes

Key observations: The dataset mixes location metadata (Locality, Longitude, Latitude), service flags (online delivery, table booking), and customer engagement metrics (rating, votes). Although a few rows at the top are from cities in the Philippines (e.g. Makati City), the dataset is dominated by Indian restaurants, especially from the Delhi NCR region.

2. Data Cleaning

Before doing any analysis I cleaned up the data. I loaded the dataset with **pd.read_csv()** and checked the structure. The shape is **9552 rows and 22 columns**.

2.1 Missing value analysis

For missing values, I found small numbers of nulls in a few columns. The numbers were small so they did not affect the analysis much. The counts per column are:

Column	Missing values
Cuisines	10
Price range	2
Aggregate rating	2
Rating color	2
Rating text	2

Column	Missing values
Votes	2
Locality	1
Locality Verbose	1
Longitude	1
Latitude	1
Average Cost for two	1
Currency	1
Has Table booking	1
Has Online delivery	1
Is delivering now	1
Switch to order menu	1
All remaining columns	0

2.2 Duplicate records

For duplicates, I used **df.duplicated()** and found **0 fully duplicated rows**, so no records needed to be dropped.

2.3 Data type changes performed

One important thing I had to fix was the **Average Cost for two** column. It was stored as text (*object* dtype) even though it contains numbers. I converted it to a numeric type using **pd.to_numeric(..., errors="coerce")** so I could do calculations on it.

Column	Before	After	Method used
Average Cost for two	object (text)	float64 (numeric)	pd.to_numeric(..., errors="coerce")

Before converting, I checked every value with **.str.isnumeric()** and found only two non-numeric entries - a NaN for *Super Loco* and a *"No"* text value - which the conversion coerced to NaN. After conversion the column has 9550 valid numeric values (9552 rows minus the 2 invalid/missing entries).

3. Data Analysis

3.1 Statistical summary - Average Cost for two

For the statistical analysis, I used **describe()** on the Average Cost for two column:

Statistic	Value	Statistic	Value
Count	9550	75th percentile (Q3)	700
Mean	1199.33	Maximum	800000
Standard deviation	16122.02	Minimum	0
25th percentile (Q1)	250	Median (50th percentile)	400

The mean, median and mode were **1199.33**, **400** and **500**. The mean is much higher than the median, which means the data is right-skewed because of a few very expensive restaurants (the maximum is 800000).

3.2 Answers to the analysis questions

Q. How many unique cities are present in the dataset?
Ans. There are **142 unique cities**.

Q. Which city has the highest number of restaurants?
Ans. **New Delhi**, with **5473 restaurants**.

Q. Which cuisine appears most frequently?
Ans. **North Indian** appears the most, in **936 restaurants**.

Q. What is the average restaurant rating?
Ans. The average rating is **2.67** out of 5.

Q. Which restaurant has the highest number of votes?
Ans. **Toit** in Bangalore, with **10934 votes**.

Q. What is the average cost for two across all restaurants?
Ans. The average is **1199.33**.

Q. Compare the average ratings of restaurants that have Online Delivery and those that do not.
Ans. Restaurants with online delivery have an average rating of **3.25**, and those without have **2.47**. So delivery restaurants are rated about **0.78** higher.

Q. Compare the average ratings of restaurants that offer Table Booking and those that do not.
Ans. Restaurants with table booking have an average rating of **3.44**, and those without have **2.56**. So booking restaurants are rated about **0.88** higher.

Q. What are the top 10 cities with the highest number of restaurants?
Ans. See the table below.

Rank	City	Number of restaurants
1	New Delhi	5473
2	Gurgaon	1118
3	Noida	1080
4	Faridabad	251
5	Ghaziabad	25
6	Ahmedabad	21
7	Bhubaneshwar	21
8	Lucknow	21
9	Guwahati	21
10	Amritsar	21

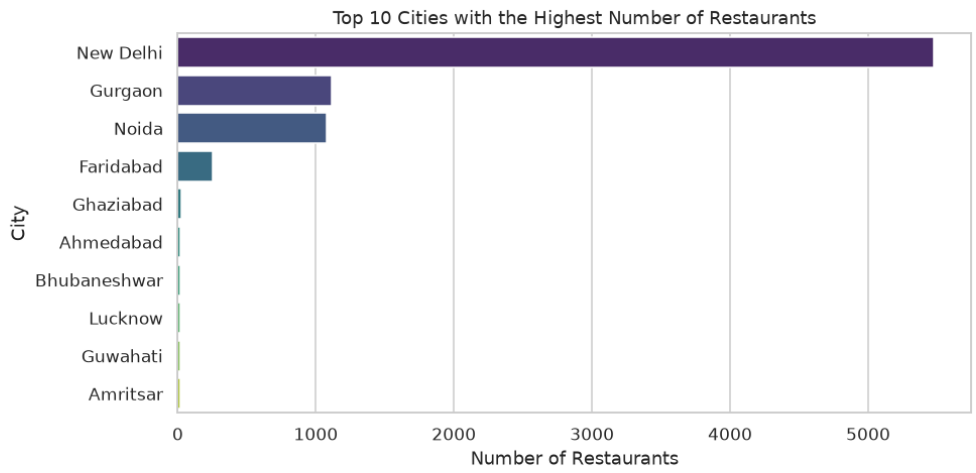
3.3 Key observations

- The average cost distribution is right-skewed: a small number of very expensive restaurants pull the mean (1199.33) far above the median (400).
- Restaurants with online delivery are rated about 0.78 higher on average than those without.
- Restaurants with table booking are rated about 0.88 higher on average than those without - an even bigger gap than online delivery.
- Restaurants are heavily concentrated in the Delhi NCR region (New Delhi, Gurgaon, Noida, Faridabad), which together make up the bulk of the dataset.

4. Data Visualizations

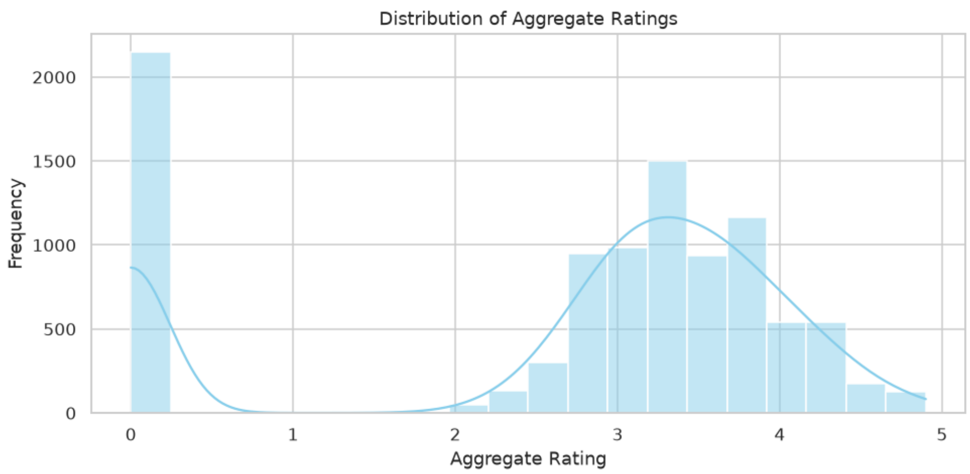
I created five charts using Matplotlib and Seaborn, all with titles and axis labels. A short interpretation is given under each chart.

Chart 1: Bar chart of the top 10 cities with the highest number of restaurants



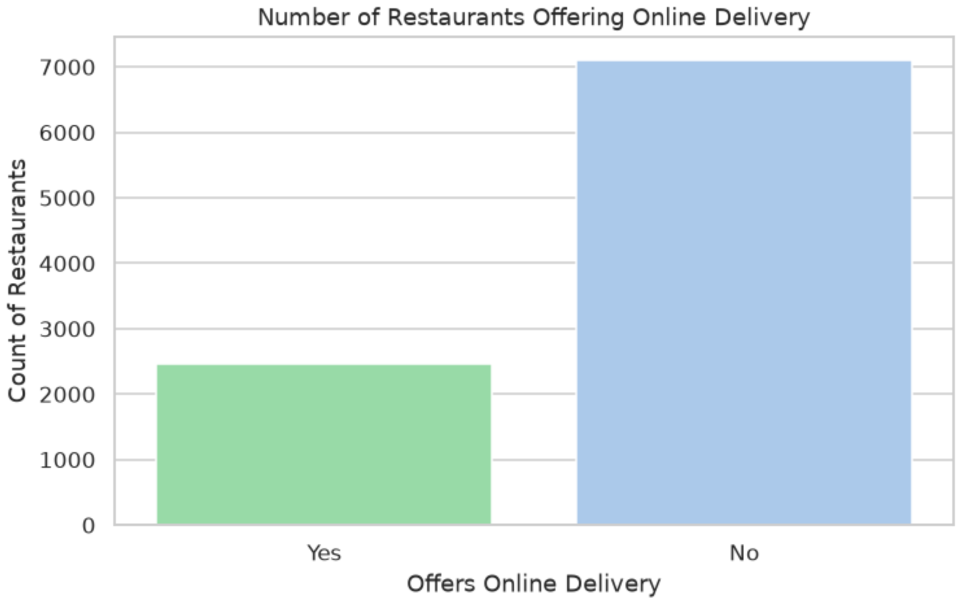
New Delhi stands out with the most restaurants, followed by Gurgaon and Noida. The restaurants are heavily concentrated in the Delhi NCR region and all other cities trail far behind.

Chart 2: Histogram of the distribution of Aggregate Ratings



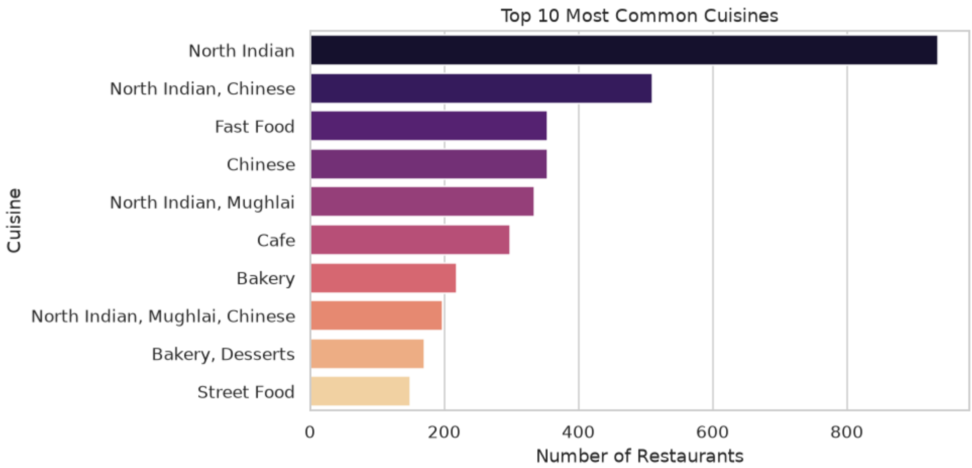
The ratings are bi-modal. Most restaurants are grouped around 3.0 to 3.5, and there is another smaller group around 4.5 and above (these are mostly restaurants that have no user ratings). Ratings near 4.0 are less common.

Chart 3: Bar chart of the number of restaurants offering Online Delivery



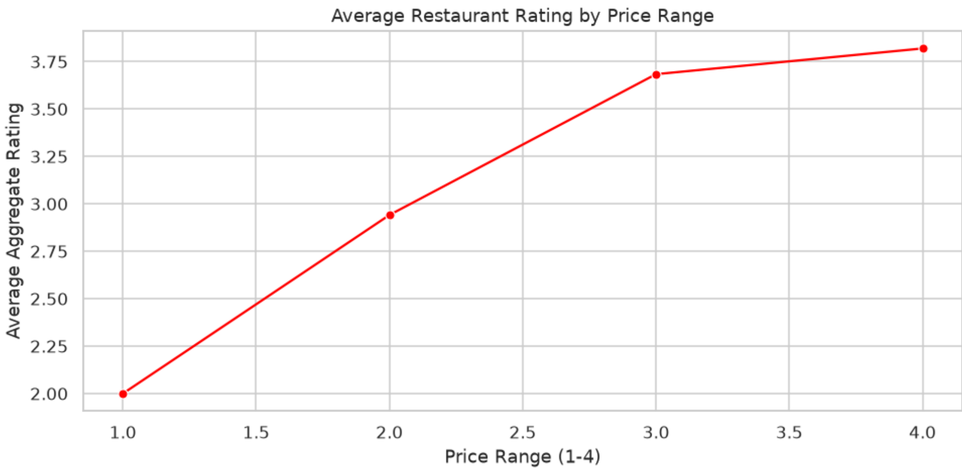
Most restaurants do not offer online delivery. Out of the dataset, **7099** restaurants say No and only **2451** say Yes. So delivery is not the main channel in this data.

Chart 4: Bar chart of the top 10 most common cuisines



North Indian is the most common cuisine with **936** restaurants, followed by the North Indian + Chinese combination with **511**. Fast Food and Chinese come next with **354** each. Multi-cuisine combos are very common, which matches how restaurant menus usually are.

Chart 5: Line chart of the average restaurant rating by Price Range



The average rating increases steadily as the price range goes from 1 to 4. The most expensive restaurants get the highest average ratings, so it looks like pricier restaurants tend to receive better reviews.

5. Insights and Conclusion

Q. Which city has the highest restaurant presence?
Ans. **New Delhi** has the highest presence with **5473 restaurants**, which makes the Delhi NCR region the biggest market in the dataset.

Q. What is the most popular cuisine?
Ans. **North Indian** is the most popular cuisine.

Q. Do restaurants with Online Delivery generally have higher ratings?
Ans. Yes. Their average rating is **3.25** compared to **2.47** for restaurants without delivery, a difference of about **0.78**.

Q. Do restaurants offering Table Booking receive better ratings?
Ans. Yes. They average **3.44** compared to **2.56** for those without, a difference of about **0.88**, which is even bigger than the delivery difference.

Q. What did I learn from analyzing this dataset?
Ans. I learned the whole EDA workflow from start to finish. I loaded a real dataset, handled a messy column where cost was stored as text, checked for missing values and duplicates, computed summary statistics, and converted each question into a grouped aggregation or a chart. The main business takeaway is that higher price range, table booking and online delivery are all linked to better ratings, so restaurants that invest in these things seem to earn more satisfied customers.

Business observations

- Higher price range, table booking and online delivery are all linked to better ratings, so restaurants that invest in these things seem to earn more satisfied customers.
- The Delhi NCR region dominates the market in this dataset - New Delhi alone accounts for over half of all restaurants (5473 out of 9552).
- Online delivery is under-utilised as a channel (only 2451 of 9552 restaurants offer it), which could be an opportunity for restaurants to expand reach.
- The restaurant market is spread over 142 cities, so any marketing or expansion strategy should be city-aware rather than one-size-fits-all.